Navbea | AI-Powered Safety & Security Solutions



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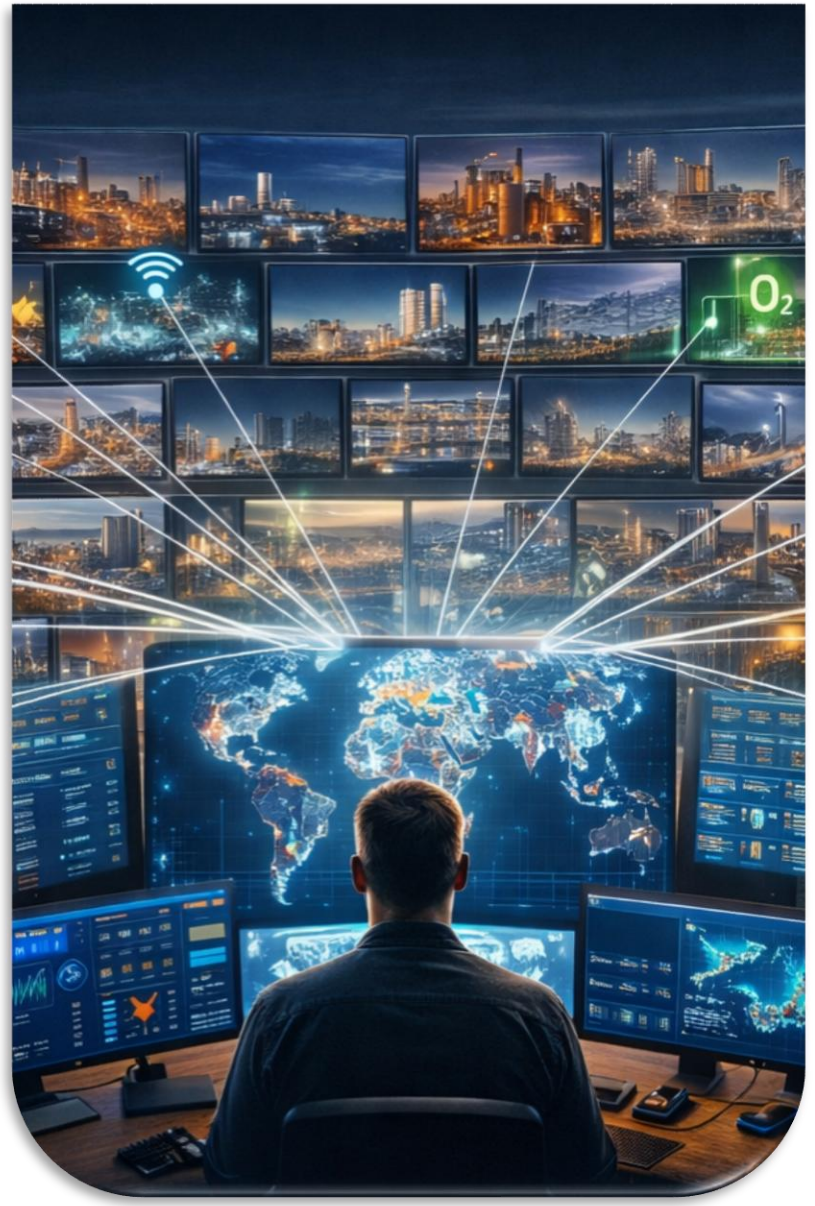
Executive Summary

- AI-driven safety and security platform designed for complex industrial environments
- Centralized multi-site management from a single command center for streamlined operations
- Real-time risk detection and rapid response capabilities to enhance workplace safety
- Proven effectiveness in heavy industry settings, ensuring reliability and scalability

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Industry Challenges in Multi-Site Operations

- Operations span dispersed plants, quarries, and terminals, complicating oversight.
- Central command visibility is limited, hindering real-time risk detection.
- Safety standards vary across sites, causing inconsistency in implementation.
- Manual and delayed reporting slows response to incidents and risk.
- Coordination challenges arise from geographic and operational diversity.



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Navbea Unified Platform Overview

Smart Sensors & Wearables

Equipped with environmental and motion sensors, wearables provide real-time monitoring of personnel safety and environmental conditions across all sites.

Fixed & Mobile Gateways

Gateways collect and transmit data from sensors and wearables, supporting both fixed installations and mobile units for flexible coverage.

AI Analytics Engine

Processes vast streams of sensor data using artificial intelligence to detect risks, predict hazards, and generate actionable safety alerts.

Central Command & Control Platform

A unified interface that consolidates data, enables multi-site management, coordinates responses, and provides comprehensive operational oversight.

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Multi-Site Management – Single Command Center

Unified Platform for All Facilities

Manage safety and security across all plants, quarries, and terminals from one centralized platform, enabling streamlined oversight and coordination.

Centralized Alarms & Workflows

Receive and respond to alarms in real time with automated workflows that ensure consistent and timely action across all sites.

Site-Level and Global KPIs

Monitor key performance indicators at both individual site and global levels to track safety performance and identify risks proactively.

Standardized Safety Operations

Implement uniform safety protocols and procedures across all locations to maintain consistent standards and improve compliance.

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Digital Mapping & Site Visualization

Indoor & Outdoor Maps

Provides detailed digital maps covering both indoor facilities and outdoor areas for comprehensive site coverage and navigation.

Plant, Quarry & Yard Visualization

Visualizes different types of operational areas including plants, quarries, and yards, allowing tailored safety monitoring for each environment.

Zone-Based Risk Layers

Implements layered risk zones on maps to identify and highlight high-risk areas for proactive safety management and hazard prevention.

Live Heatmaps

Displays real-time heatmaps to track activity levels, risk concentrations, and personnel movement for dynamic situational awareness.

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Personnel Tracking

Real-Time Personnel Localization (RTLS)

01 Indoor & Outdoor Positioning

Accurate real-time tracking of personnel both inside buildings and across outdoor site areas ensures continuous monitoring and safety compliance.

02 Zone Entry/Exit Tracking

Automatically records when personnel enter or leave designated safety zones, helping to enforce site access policies and monitor movement patterns.

03 Unauthorized Access Detection

Alerts triggered by personnel entering restricted or hazardous areas without proper clearance, enabling immediate security intervention.

04 Time-in-Zone Analysis

Detailed reports on how long individuals remain in specific zones provide insights for operational efficiency and risk management.

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AI Safety – Fall & Immobility Detection

- Continuous motion analysis identifies abnormal movement patterns indicating falls or immobility.
- Automatic alerts are sent immediately to the central command center for quick intervention.
- Escalation procedures activate if the initial alert receives no response, ensuring no incident is overlooked.
- This system enhances worker safety by enabling faster emergency response and reducing risk of delayed assistance.



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AI Safety

AI Safety – Gas & Environmental Intelligence

- Sensors monitor critical gases including carbon monoxide (CO) and oxygen (O_2) levels continuously.
- Temperature sensors track environmental conditions that can affect safety risks.
- AI correlates sensor data with personnel location and motion for precise risk assessment.
- Predictive analytics generate early warnings based on detected gas levels and environmental trends.
- Alerts enable prompt action to prevent incidents related to toxic gases and unsafe conditions.

Emergency & Muster Management

- Live evacuation monitoring provides real-time updates on personnel movement during emergencies.
- Digital headcount feature automatically tracks the number of people present at each site or muster point.
- Missing personnel detection alerts safety teams immediately if expected personnel are unaccounted for.
- Integrated communication systems enable swift coordination and response during evacuations.

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Security

AI-Powered Security & Compliance

- Restricted area enforcement with real-time access control to prevent unauthorized entry.
- Comprehensive tracking of contractors and visitors for enhanced site security.
- Automated digital records for audits, ensuring compliance transparency and accountability.
- Integration with safety systems to trigger alerts on security breaches.
- Supports regulatory compliance with detailed, time-stamped activity logs.

Business & ROI Impact

Reduced Incidents & Downtime

AI-powered detection and predictive alerts minimize accidents and operational interruptions, significantly lowering incident rates across all sites.

Faster Emergency Response

Real-time alerts and centralized command enable rapid mobilization of emergency teams, reducing response times and improving outcomes.

Consistent Global Standards

Standardized workflows and centralized monitoring enforce uniform safety protocols across multiple sites, ensuring compliance and reducing risks.

Typical ROI: 6–18 Months

Investment in Navbea's platform typically recoups costs within 6 to 18 months through operational savings and incident reduction benefits.

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Advantages

Why Navbea for Heidelberg Materials

Designed for Heavy Industry

Navbea’s platform is engineered to withstand harsh industrial environments, supporting operations in plants, quarries, and terminals with robust AI safety and security solutions tailored to heavy industry demands.

Scalable Across Hundreds of Sites

The solution can easily scale from a single facility to hundreds of sites globally, enabling centralized management and consistent safety standards across all locations.

Hardware-Agnostic & Future-Proof

Navbea supports integration with a wide range of existing sensors and devices, ensuring flexibility and protection of prior investments while adapting to future technological advances.

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Implementation

Next Steps

Site Mapping & Risk Assessment

Conduct detailed mapping of all relevant sites including plants, quarries, and yards to identify critical risk areas and infrastructure needs.

- Digital site maps
- Risk zone identification
- Infrastructure assessment report

Pilot at Selected Facility

Deploy the Navbea platform in a controlled environment at one facility to validate system performance, user experience, and integration with existing operations.

- Pilot deployment plan
- Performance evaluation report
- User feedback collection

ROI Modeling

Analyze pilot data to model potential return on investment, including projected incident reduction, operational efficiencies, and emergency response improvements.

- Cost-benefit analysis
- ROI projection report
- Stakeholder presentation

Phased Global Rollout

Develop and execute a phased deployment plan to scale the platform across all sites, ensuring standardized safety procedures and centralized management.

- Rollout schedule
- Training programs
- Standard operating procedures
- Central command integration

Thank you

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